RELATED RATES

BLAKE FARMAN

Lafayette College

Name: Solutions

1. Gas is escaping a spherical balloon at the rate of 4 cm^3 per minute. How fast is the surface area shrinking when the radius is 24 cm? For a sphere, $V = \frac{4}{3}\pi r^3$ and $S = 4\pi r^2$ where V is volume, S is surface area and r is the radius of the balloon.

$$\frac{dS}{dt} = 4\pi\left(2r\frac{dr}{dt}\right) = 8\pi r\frac{dr}{dt}$$

So to find $\frac{dS}{dt}$, we need $\frac{dr}{dt}$:

$$\frac{dV}{dt} = \frac{4}{3}\pi\left(3r^2\frac{dr}{dt}\right) = 4\pi r^2\frac{dr}{dt}$$

$$\Rightarrow \frac{dr}{dt} = \frac{1}{4\pi r^2}\frac{dV}{dt}$$

$$\Rightarrow \frac{dS}{dt} = 8\pi r\left(\frac{1}{4\pi r^2}\frac{dV}{dt}\right)$$

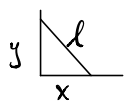
$$= \frac{2}{r}\frac{dV}{dt}$$

$$= \frac{2}{24}(-4)$$

$$= \frac{-2}{6}$$

$$= \boxed{-\frac{1}{3}\ cm^2/min.}$$

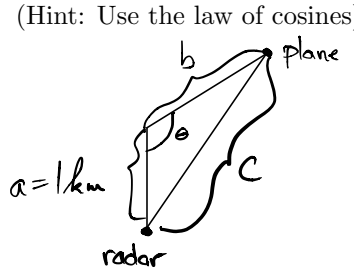
2. The top of a ladder slides down a vertical wall at a rate of 0.15 meters/second. At the moment when the bottom of the ladder is 3 meters from the wall, it slides away from the wall at a rate of 0.2 meters/second. How long is the ladder?



$\ell^2 = y^2 + x^2 \Rightarrow \ell = \sqrt{y^2+x^2} = \sqrt{y^2+9}$, so we need to find the value of y when $x=3$. We know $\frac{dy}{dt} = \frac{15}{100} = \frac{3}{20}$ and $\frac{dx}{dt} = \frac{2}{10} = \frac{1}{5}$ when $x=3$, so we apply $\frac{d}{dt}$ to both sides of $\ell^2 = y^2+x^2$ to get

$$0 = 2y\frac{dy}{dt} + 2x\frac{dx}{dt} \quad (\ell \text{ is a constant})$$
$$= 2y\left(\frac{-3}{20}\right) + 2(3)\left(\frac{1}{5}\right)$$
$$\Rightarrow \frac{3y}{10} = \frac{6}{5}$$
$$\Rightarrow y = \frac{6}{5}\left(\frac{10}{3}\right) = \left(\frac{6}{3}\right)\left(\frac{10}{5}\right) = 2^2 = 4 \Rightarrow \ell = \sqrt{16+9} = \sqrt{25} = \boxed{5\text{ m}}$$

3. A plane flying with a constant speed of 300 kilometers/hour passes over a ground radar station at an altitude of 1 kilometer and climbs at an angle of 30°. At what rate is the distance from the plane to the radar station increasing a minute later?
(Hint: Use the law of cosines)



We are given $\frac{db}{dt} = 300$ km/hr and $\theta = 30° + 90° = \frac{3\pi}{2}$ rads

We are asked to find $\frac{dc}{dt}$.

The Law of Cosines says

$$c^2 = a^2 + b^2 - 2ab\cos(\theta)$$
$$= 1 + b^2 - 2b\left(\frac{-1}{2}\right)$$
$$= 1 + b + b^2.$$

By applying $\frac{d}{dt}$ to both sides we get

$$2c\frac{dc}{dt} = 0 + \frac{db}{dt} + 2b\frac{db}{dt}$$
$$\Rightarrow \frac{dc}{dt} = \frac{(1+2b)\frac{db}{dt}}{2c}$$

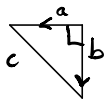
We have $b = 300\left(\frac{1}{60}\right) = 5$ km and

$$c = \sqrt{1+5+5^2} = \sqrt{31}$$

so

$$\frac{dc}{dt} = \frac{(1+2(5))(300)}{\sqrt{31}} = \frac{11(300)}{\sqrt{31}} = \boxed{\frac{3300}{\sqrt{31}}} \approx 296 \text{ km/hr}$$

4. Two cars start moving from the same pont. One travels south at 60 mi/h and the other travels west at 25 mi/h. At what rate is the distance between the cars increasing two hours later?



Given $\frac{da}{dt} = 25$ mph and $\frac{db}{dt} = 60$ mph, we want to find $\frac{dc}{dt}$.

$a = 25t$, $b = 60t$

$\Rightarrow c^2 = 25^2t^2 + 60^2t^2 = (25^2 + 60^2)t^2 = 5^2(5^2 + 12^2)t^2 = 5^2(169)\,t^2$

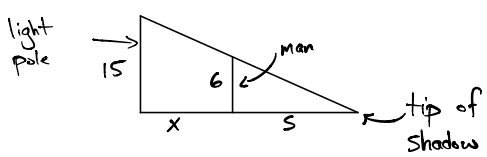
$= 5^2 13^2 t^2 = (65t)^2$

$\Rightarrow c = 65t$

$\Rightarrow \boxed{\frac{dc}{dt} = 65 \text{ mph}}$

5. A street light is mounted at the top of a 15-ft-tall pole. A man 6 ft tall walks away from the pole with a speed of 5 ft/s along a straight path. How fast is the tip of his shadow moving when he is 40 ft from the pole?

(Hint: The length of the shadow is measured from the person to the tip of the shadow; the rate at which the tip of the shadow is moving is measured from the pole to the tip of the shadow.)



By similar triangles

$\frac{15}{x+s} = \frac{6}{s}$

$\Rightarrow 15s = 6(x+s) = 6x + 6s$

$\Rightarrow 6x = 15s - 6s = 9s$

$\Rightarrow \frac{ds}{dt} = \frac{6}{9}\frac{dx}{dt} = \frac{2}{3}\frac{dx}{dt}$

By the hint we need to find

$$\frac{d}{dt}(x+s) = \frac{dx}{dt} + \frac{ds}{dt} = \frac{dx}{dt} + \frac{2}{3}\frac{dx}{dt} = \frac{dx}{dt}\left(\frac{3}{3} + \frac{2}{3}\right) = \frac{5}{3}\frac{dx}{dt}$$

We are given that $\frac{dx}{dt} = 5$ when $x = 40$, so the tip of the shadow is moving at a rate of

$$5\left(\frac{5}{3}\right) = \boxed{\frac{25}{3} \text{ ft/s}}$$